

School of Computer Science Engineering and Technology

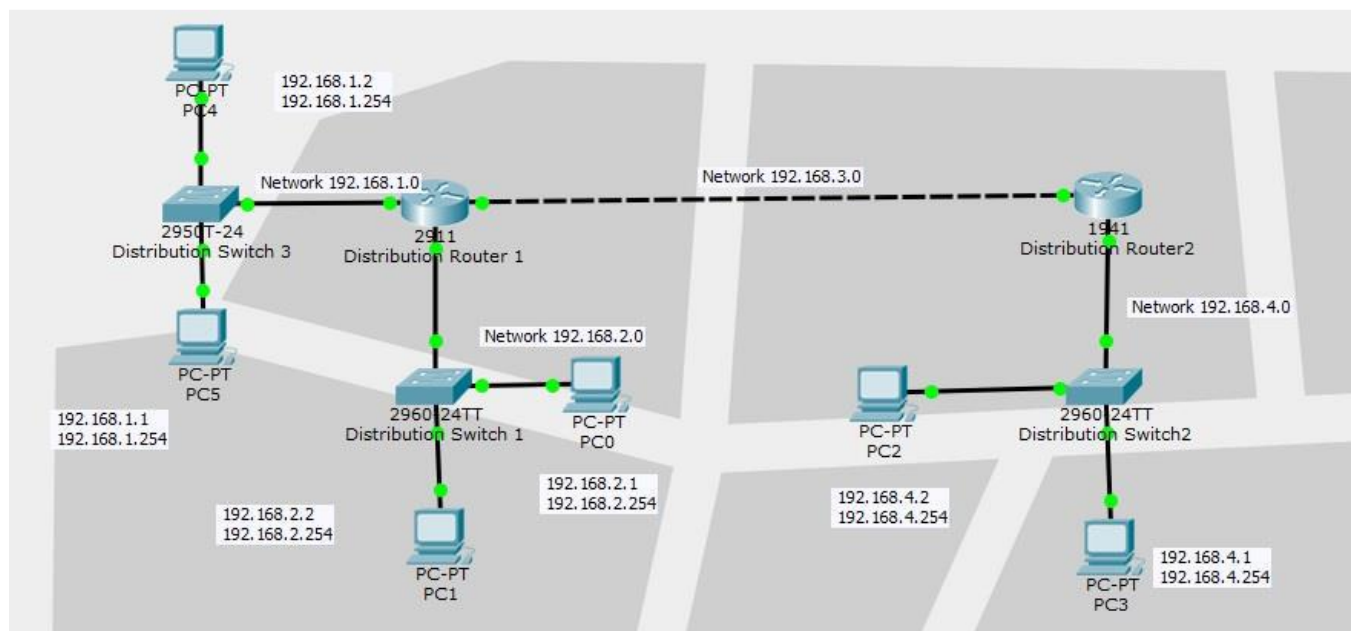
Course- B. Tech	Type- Core
Course Code- CSET207	Course Name- COMPUTER NETWORKS
Year- 2025-26	Semester- Even
Date- 9-13/02/2026	Batch- 2024-2028

CO-Mapping

	CO1	CO2	CO3
Q1	✓		
Q2	✓		
Q3	✓		
Q4	✓	✓	

Objective – Understanding Routing Information Protocol (RIP).

Task 1: Design the following topology in Packet Tracer. The Network classification is annotated.:



Task 2: Configure Router Interfaces.

Distribution Router 1

```

Router>enable
Router#configure terminal
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.254 255.255.255.0
  
```

School of Computer Science Engineering and Technology

```
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.2.254 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
```

Distribution Router 2

```
Router>enable
Router#configure terminal
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#ip address 192.168.3.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.4.254 255.255.255.0
Router(config-if)#no shutdown
```

Task 3: Configure Routing Information Protocol.

Routing Information Protocol (RIP) is a commonly used routing protocol in small to medium TCP/IP networks. It is a stable protocol that uses a distance-vector algorithm to calculate routes. Routing Information Protocol (RIP) uses hop count as the metric to rate the value of different routes. The hop count is the number of devices that can be traversed in a route. A directly connected network has a metric of zero; an unreachable network has a metric of 16. This limited metric range makes RIP unsuitable for large networks.

The Routing Information Protocol (RIP) uses broadcast UDP data packets to exchange routing information. Cisco software sends routing information updates every 30 seconds, which is termed advertising. If a device does not receive an update from another device for 180 seconds or more, the receiving device marks the routes served by the no updating device as unusable. If there is still no update after 240 seconds, the device removes all routing table entries for the no updating device. A device that is running RIP can receive a default network via an update from another device that is running RIP, or the device can source the default network using RIP. In both cases, the default network is advertised through RIP to other RIP neighbours.

RIP Routing Updates

The Routing Information Protocol (RIP) sends routing-update messages at regular intervals and when the network topology changes. When a device receives a RIP routing update that includes changes to an entry, the device updates its routing table to reflect the new route. The metric value for the path is increased by 1, and the sender is indicated as the next hop. RIP devices maintain only the best route (the route with the lowest metric value) to a destination. After updating its routing table, the device immediately begins transmitting RIP routing updates to inform other network devices of the change. These updates are sent independently of the regularly scheduled updates that RIP devices send.

RIP Routing Metric

The Routing Information Protocol (RIP) uses a single routing metric to measure the distance between the source and the destination network. Each hop in a path from the source to the destination is assigned a hop-count value, which is typically 1. When a device receives a routing update that contains a new or changed destination network entry, the device adds 1 to the metric value indicated in the update and enters the network in the routing table. The IP address of the sender is used as the next hop. If an interface network is not specified in the routing table, it will not be advertised in any RIP update.

School of Computer Science Engineering and Technology

Distribution Router 1

Router(config)#router rip

Router(config-router)#network 192.168.1.0

Router(config-router)#network 192.168.2.0 Router(config-router)#network 192.168.3.0

Distribution Router 2

Router(config)#router rip

Router(config-router)#network 192.168.3.0

Router(config-router)#network 192.168.4.0

Task 4: Verify Configuration

Router# show ip rip database

Router# show ip rip route

Task 5: Supplementary Exercise

Using the same topology implement RIP Version 2, practice active and passive interfaces and route summarization.